Ch 3.1: More Linear Regression

Lecture 5 - CMSE 381

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Mon, Sep 9, 2024

Announcements

Last time:

- Started 3.1 - Single linear regression

Announcements:

- Office Hours
- Homework #1 Grading TBD
- Homework #2 Due Sun, Sep 15
- Quiz 2 - Friday

Covered in this lecture

- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared

Section 1

Last time

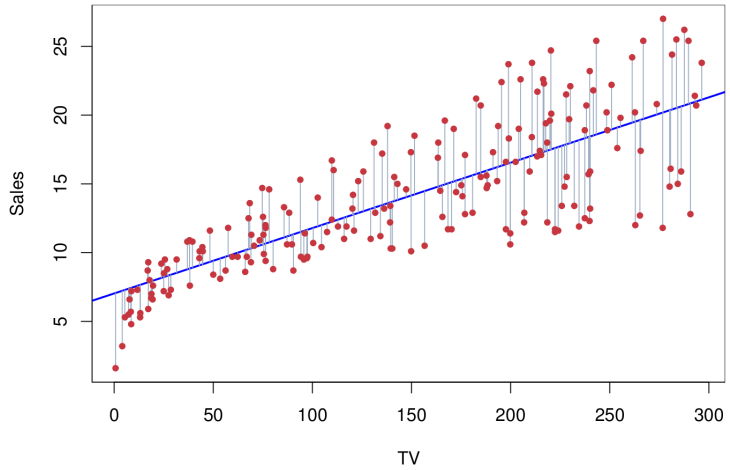
Setup

- Predict Y on a single predictor variable X

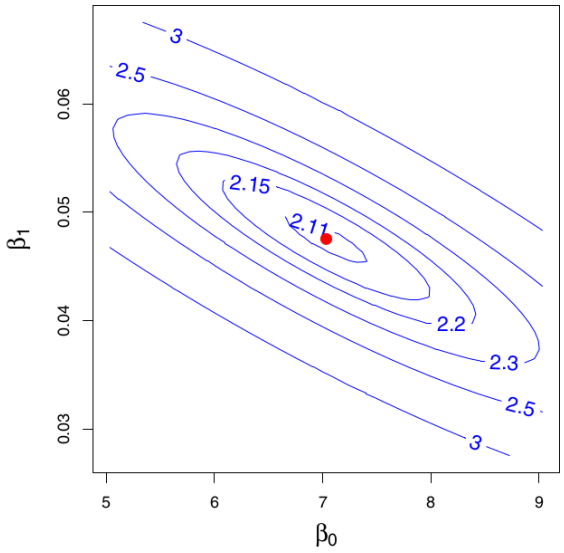
$$Y \approx \beta_0 + \beta_1 X$$

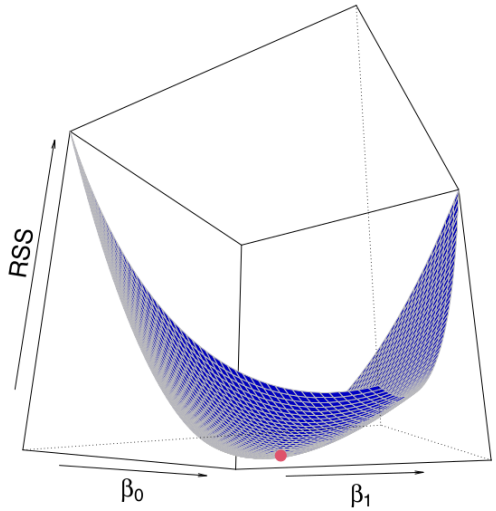
- "$\approx$" .... "is approximately modeled as"

- Given $(x_1, y_1), \cdots, (x_n, y_n)$
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be prediction for Y on ith value of X.
- $e_i = y_i - \hat{y}_i$ is the ith residual



Least squares criterion: RSS





Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Least squares criterion

Find β_0 and β_1 that minimize the RSS.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Section 2

Assessing Coefficient Estimate Accuracy

Bias in estimation

Analogy with mean

- Assume a true value μ^*
- An estimate from training data $\hat{\mu}$
- The estimate is unbiased if $E(\hat{\mu} = \mu^*)$

- Sample mean is unbiased for population mean:

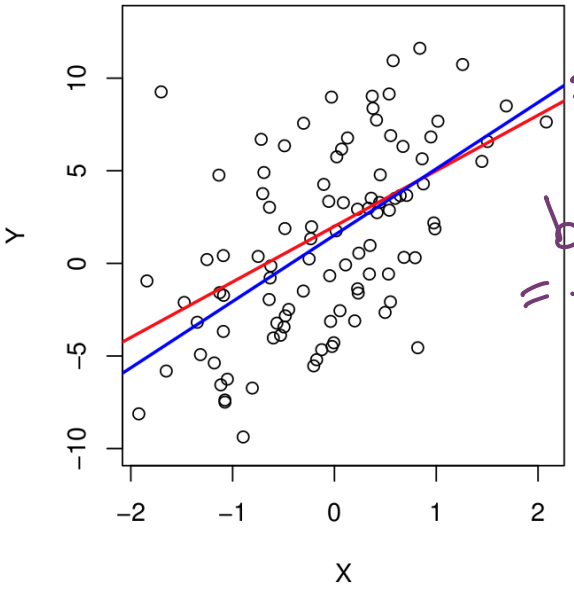
$$E(\hat{\mu}) = E\left(\frac{1}{n}\sum_i X_i\right) = \mu$$

- Standard variance estimate is biased

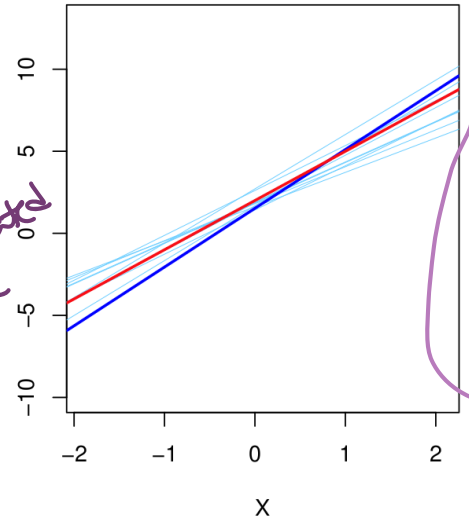
$$E(\hat{\sigma}^2) = E\left[\frac{1}{n}\sum_i (X_i - \overline{X})^2\right] \neq \sigma^2$$

Linear regression is unbiased

red = true line $f(x) = 2 + 3x$



blue = estimated line



- Sample 100 data pts from $Y = 2 + 3x + \varepsilon$
- ε drawn from normal dist mean 0
- Get my lin reg line
- Repeat $\boxed{\text{this}}$ 10 times

The lin. reg lines are light blue

Variance in estimation

Continuing analogy with mean

- True value μ^*
- Estimate from training data $\hat{\mu}$
- Variance of sample mean
  $\mathrm{Var}(\hat{\mu}) = \mathrm{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n}$ ← variance of distribution
  ← # data pts
  ↳ standard error

Variance of linear regression estimates

- Variance of linear regression estimates:

Fix this!!!

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

Residual sum of squares

Want to estimate β_0^*, β_1^*

(example before

$Y = 2 + 3X + \varepsilon$

$\beta_0^* = 2 \quad \beta_1^* = 3$)

estimate of $\sigma^2 = (RSE)^2$

Coding group work

Run the section titled "Simulating data"

Confidence Interval

The version for β_0 is the same

The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

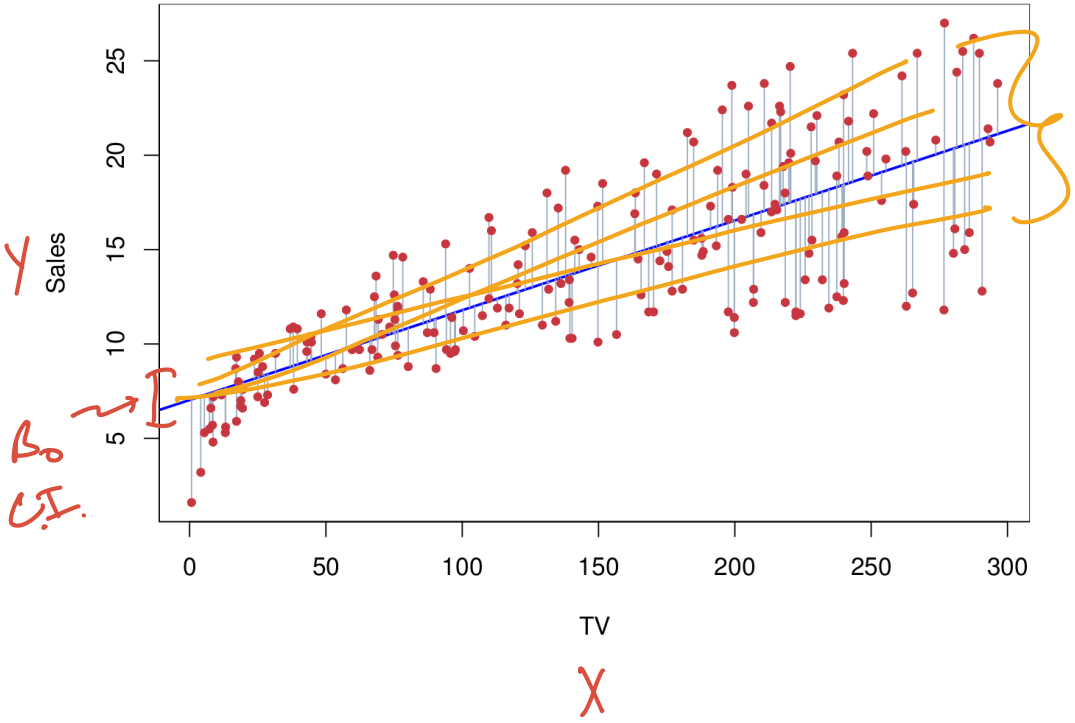
coef from my data's lin. reg.

Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.

CI in Advertising data



lines w/ slope in β_1 conf. int.

For the advertising data set, the 95% CIs are:

- β_1 :: $[0.042, 0.053]$
- β_0 :: $[6.130, 7.935]$

Intercept

Hypothesis testing

"Moral"

H_0: There is no relationship between X and Y

H_a: There is some relationship between X and Y

no relationship

$Y = (\beta_0 + \beta_1 X) + \text{noise}$

$$\boxed{\begin{aligned} H_0 &: \beta_1 = 0 \\ H_a &: \beta_1 \neq 0 \end{aligned}}$$

Real way to think about this

Test statistic and p-value

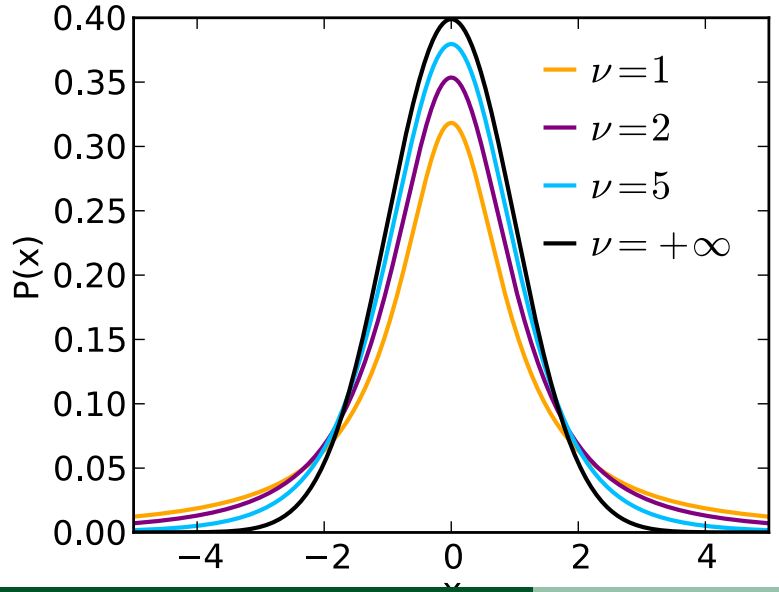
Test statistic:

$$\boxed{t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}}$$

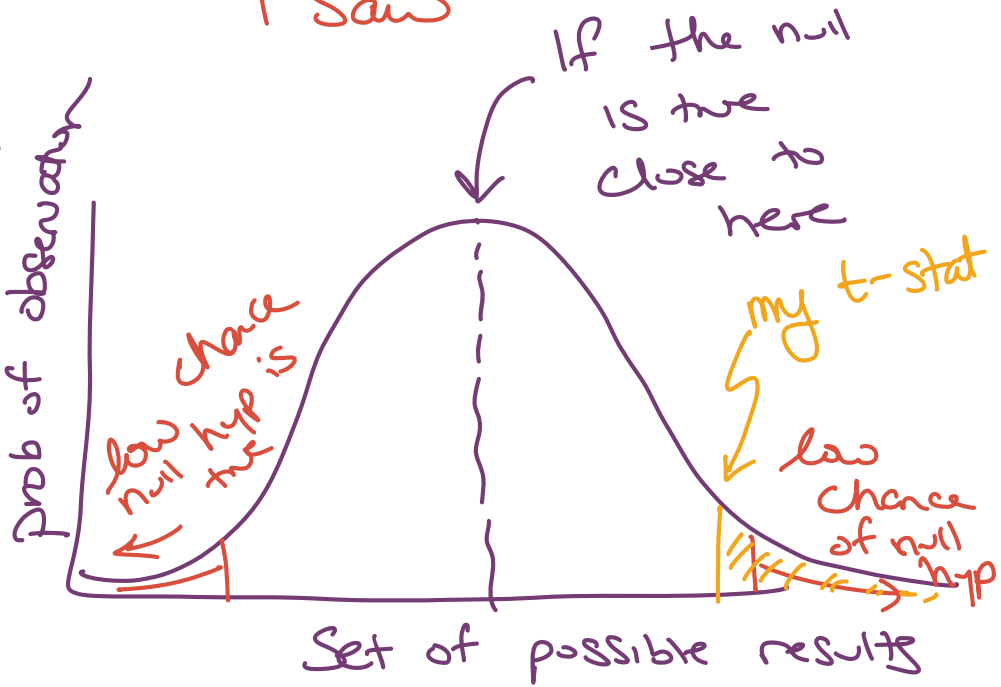
calc from data

t-distribution with $n - 2$ degrees of freedom

# of data pts

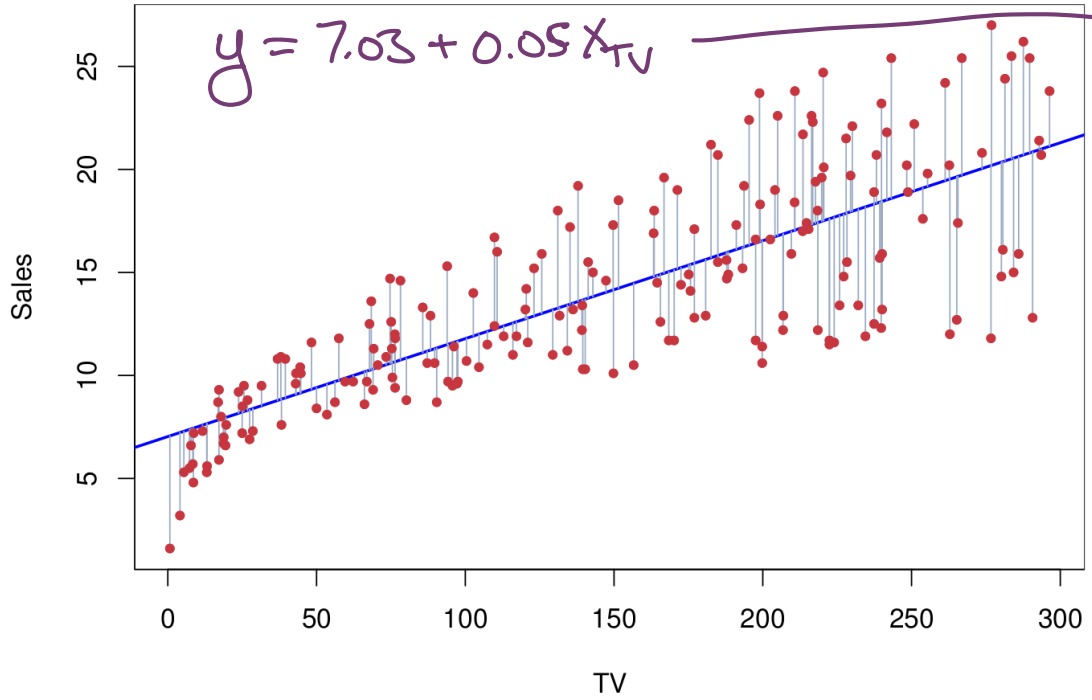


p-val: prob of seeing an event more contradictory to the null hyp than what I saw



Advertising example

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001



Small p-value means reject the null

$\beta_1 = 0$

Assessing the accuracy of the module: RSE

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$
$$= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}$$

Quantify the extent to which the model fits the data (all of this!)

- an estimate of the std dev of ε $= \sqrt{Var(\varepsilon)}$
- Avg amt that the response will deviate from the true reg line

Assessing the accuracy of the module: R^2

R squared:

$\sum_i (y_i - \hat{y}_i)^2$

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where total sum of squares is

$$TSS = \sum_i (y_i - \bar{y})^2$$

Total Sum of squares

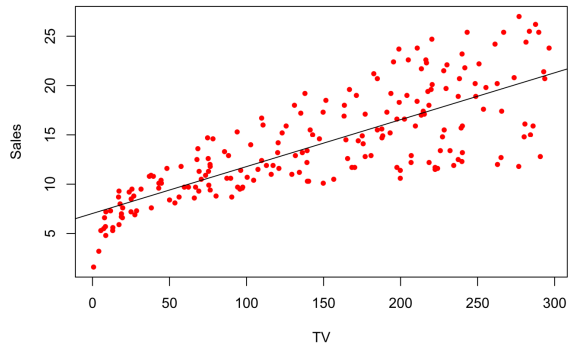
- TSS is the total variance in the response Y before taking X into account
  Variability before regression
- RSS
  Variability after the regression
- R^2 is the proportion of variability explained w/ X

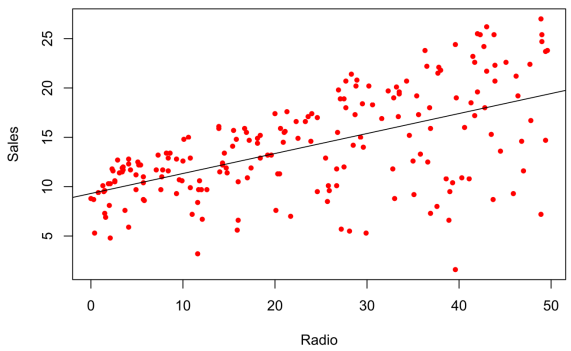
not much explained by X

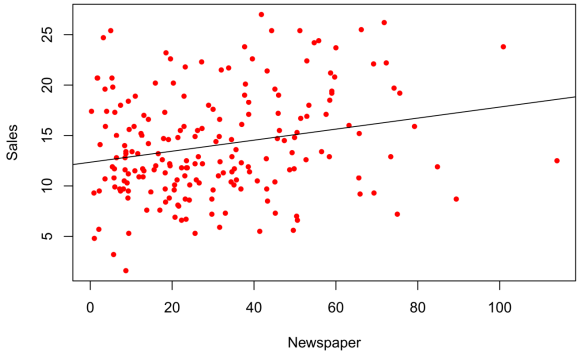
lots explained by X

0 — 1

Advertising example







$R^2 = 0.61$

$R^2 = 0.33$

$R^2 = 0.05$

Coding group work

Run the section titled "Assessing Coefficient Estimate Accuracy"

Next time

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1/3.2	
6	Wed	9/11	Even more linear regression	3.2.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due

Announcements

- Homework 2
  - Due Sun, Sep 15